

Practical Guide to Gamma Greeks

Quant Insider

June 6, 2025

Abstract

This document explains how to compute and use Gamma and related second- and third-order Greeks in real-world option trading and risk management. Included are:

- How to calculate standard Gamma (Γ) and interpret it in practice
- Approximations for fast evaluation (Saddle Gamma)
- Normalizing Gamma (GammaP) for position sizing
- Using Gamma symmetry for skew analysis
- Evaluating sensitivity of Gamma to volatility (VommaGamma)
- Using Speed ($\partial\Gamma/\partial S$) for dynamic hedging
- Using Color ($\partial\Gamma/\partial T$) to understand time decay of risk

Throughout, we show numerical examples and discuss how traders and risk managers incorporate these metrics into daily workflows.

Contents

1	Introduction	3
2	Standard Gamma: Calculation and Example	3
2.1	Formula	3
2.2	Numerical Example	3
2.3	Practical Notes	4
3	Saddle Gamma: Fast Approximation	4
3.1	Motivation	4
3.2	Saddle Gamma Formula (Lognormal Case)	4
3.3	When to Use	4
4	Percentage Gamma (GammaP) for Position Sizing	5
4.1	Definition and Interpretation	5
4.2	Example and Use	5
4.3	Risk Limits	5
5	Gamma Symmetry: Skew Analysis	5
5.1	Put-Call Symmetry	5
5.2	Application	5
6	VommaGamma: Sensitivity of Gamma to Volatility	5
6.1	Formula and Calculation	5
6.2	Numerical Example	6
6.3	Practical Notes	6

7	Speed: How Gamma Changes with Spot	6
7.1	Formula and Interpretation	6
7.2	Numerical Example	6
7.3	Practical Implications	6
8	Color: Gamma's Time Decay	6
8.1	Formula	6
8.2	Numerical Example	7
8.3	Use Cases	7
9	Implementing in a Risk System	7
9.1	Workflow	7
9.2	Sample Python Pseudocode	7
10	Summary and Best Practices	8

1 Introduction

Gamma is a critical measure for understanding how an option's Delta changes with small moves in the underlying. In practice, high Gamma positions require frequent rebalancing to remain hedged. This guide provides step-by-step formulas, numerical examples, and notes on how to integrate Gamma-related Greeks into trading systems and risk reports.

2 Standard Gamma: Calculation and Example

2.1 Formula

Under Black-Scholes, the Gamma for a European call or put is:

$$\Gamma = \frac{e^{-bT}}{S\sigma\sqrt{2\pi T}} \exp\left(-\frac{d_1^2}{2}\right),$$

where

$$d_1 = \frac{\ln\left(\frac{S}{X}\right) + (b + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

and

- S : current spot price of the underlying
- X : option strike
- T : time to maturity (in years)
- b : cost-of-carry (for stocks, typically $r - q$, where q is dividend yield)
- σ : implied volatility (annualized)

2.2 Numerical Example

Suppose:

- Underlying $S = 100$
- Strike $X = 100$
- Time to maturity $T = 30/365 \approx 0.0822$ years (30-day option)
- Risk-free rate $r = 5\%$ annual; no dividends, so $b = r = 0.05$
- Implied volatility $\sigma = 25\%$

First compute:

$$d_1 = \frac{\ln(100/100) + (0.05 + 0.5 \times 0.25^2) \times 0.0822}{0.25\sqrt{0.0822}} = \frac{(0.05 + 0.03125) \times 0.0822}{0.0717} \approx 0.092.$$

Then:

$$\Gamma = \frac{e^{-0.05 \times 0.0822}}{100 \times 0.25 \times \sqrt{2\pi \times 0.0822}} \exp(-0.092^2/2).$$

Numerically:

- $e^{-0.05 \times 0.0822} \approx 0.9959$
- $\sqrt{2\pi \times 0.0822} \approx 1.279$

- $\exp(-0.092^2/2) \approx 0.9958$

So

$$\Gamma \approx \frac{0.9959}{100 \times 0.25 \times 1.279} \times 0.9958 = \frac{0.9917}{31.98} \approx 0.0310.$$

Gamma is 0.0310 per \$1 move in S . A \$1 move in the underlying causes Delta to shift by about 0.031.

2.3 Practical Notes

- **Hedging Frequency:** With Gamma = 0.031, a \$1 move in the stock changes Delta by 3.1%. For a \$1 million notional position, daily price swings of \$1 require adjusting Delta by \$31,000 of the underlying to stay hedged.
- **Monitoring:** Traders track Gamma not just at spot but across a grid of strikes and maturities (a "Gamma surface") to see where risk is concentrated.
- **Real-Time Alerts:** Set thresholds for Gamma changes; alerts notify traders when Gamma exceeds risk limits.

3 Saddle Gamma: Fast Approximation

3.1 Motivation

For portfolios with thousands of option positions, computing Black-Scholes Gamma for each can be time-consuming. Saddlepoint approximations offer a faster way to estimate Gamma when extreme moves or non-lognormal features matter.

3.2 Saddle Gamma Formula (Lognormal Case)

In Black-Scholes, the cumulant-generating function is $K(q) = bq + \frac{1}{2}\sigma^2 q^2$. The saddlepoint q^* solves:

$$K'(q^*) = b + \sigma^2 q^* = \frac{\ln\left(\frac{Xe^{-bT}}{S}\right)}{T},$$

so

$$q^* = \frac{1}{\sigma^2} \left(\frac{\ln(X/S) + bT}{T} - b \right).$$

Substitute into:

$$\Gamma_{\text{saddle}} \approx \frac{e^{-bT}}{S} \sqrt{\frac{1}{2\pi T(\sigma^2)}} \exp\left(T(K(q^*) - q^* K'(q^*))\right).$$

Pre-built libraries (in Python or C++) handle these calculations once parameters are specified.

3.3 When to Use

- **Short-Dated Options:** As $T \rightarrow 0$, Gamma spikes; saddlepoint avoids numerical instabilities.
- **Heavy-Tailed Models:** For fat-tail returns (e.g., jump-diffusion), saddlepoint captures tail behavior more accurately.
- **Speed:** Reduces CPU time in risk systems recalculating Greeks for large portfolios.

4 Percentage Gamma (GammaP) for Position Sizing

4.1 Definition and Interpretation

Percentage Gamma normalizes absolute Gamma by the underlying price:

$$\Gamma_P = 100 \times \frac{\Gamma}{S},$$

measured as basis points of Delta per 1% move in the underlying. Traders use GammaP to compare risk across options on different underlyings.

4.2 Example and Use

Continuing the previous example with $S = 100$, $\text{Gamma} = 0.0310$:

$$\Gamma_P = 100 \times \frac{0.0310}{100} = 0.031\% \text{ per } 1\% \text{ move.}$$

If a portfolio has \$200,000 in option Delta notional at that strike, a 1% move changes Delta by 0.031% of \$200,000 = \$62. This helps budget hedging costs.

4.3 Risk Limits

Institutions set limits on aggregated GammaP across all options to cap the total Delta shift for a given market move.

5 Gamma Symmetry: Skew Analysis

5.1 Put-Call Symmetry

Gamma symmetry indicates call Gamma at one strike equals put Gamma at a mirrored strike:

$$\Gamma_{\text{call}}(K) = \Gamma_{\text{put}}\left(\frac{F^2}{K}\right),$$

where forward $F = Se^{(b-r)T}$. Traders use this to spot skew: if put Gammas at low strikes exceed call Gammas at mirrored strikes, the market is skewed.

5.2 Application

- **Vol Surface Construction:** Enforce Gamma symmetry when interpolating to ensure no-arbitrage.
- **Skew Monitoring:** Compare implied volatilities at K and F^2/K ; deviations signal directional bias or demand imbalances.

6 VommaGamma: Sensitivity of Gamma to Volatility

6.1 Formula and Calculation

VommaGamma measures how Gamma changes as implied volatility shifts:

$$\frac{\partial \Gamma}{\partial \sigma} = \Gamma \frac{d_1 d_2 - 1}{\sigma}.$$

6.2 Numerical Example

Using $d_1 \approx 0.092$, $d_2 = 0.0203$, $\Gamma = 0.0310$:

$$\frac{\partial \Gamma}{\partial \sigma} = 0.0310 \times \frac{0.092 \times 0.0203 - 1}{0.25} \approx -0.1238.$$

A 1% absolute increase in volatility reduces Gamma by about 0.00124.

6.3 Practical Notes

- **Vol-of-Vol Risk:** Positions with large VommaGamma are sensitive to volatility shifts. Hedge by trading Vega options.
- **Risk Reports:** Include VommaGamma exposure to assess how volatility surface moves affect hedging.

7 Speed: How Gamma Changes with Spot

7.1 Formula and Interpretation

Speed is the third derivative $\partial^3 C / \partial S^3$:

$$\text{Speed} = -\frac{e^{-bT}}{S^2 \sigma \sqrt{2\pi T}} \exp\left(-\frac{d_1^2}{2}\right) (2 + d_1 \sqrt{T} - d_1^2).$$

A negative Speed means Gamma decreases as spot moves away from at-the-money.

7.2 Numerical Example

Using $d_1 = 0.092$, $T = 0.0822$, $\sigma = 0.25$, $S = 100$, $b = 0.05$:

$$\begin{aligned} 2 + d_1 \sqrt{T} - d_1^2 &= 2 + 0.092 \times 0.2867 - 0.092^2 \approx 2.0175. \\ \text{Speed} &= -\frac{0.9959}{100^2 \times 0.25 \times 1.279} \times 0.9958 \times 2.0175 \\ &\approx -\frac{2.002}{3197.5} \approx -0.000626. \end{aligned}$$

A \$0.10 move in spot changes Gamma by approximately -0.0000626 .

7.3 Practical Implications

- **Dynamic Hedging:** Use Speed to estimate additional shares or futures to trade when spot moves a fraction, without recomputing full Gamma.
- **Cost Estimates:** Estimate transaction costs for small hedge adjustments.

8 Color: Gamma's Time Decay

8.1 Formula

Color describes $\partial \Gamma / \partial T$:

$$\text{Color} = \Gamma \left[b - \frac{1 + d_1 d_2}{2T} \right].$$

Negative Color indicates Gamma decays as time passes.

8.2 Numerical Example

With $\Gamma = 0.0310$, $b = 0.05$, $d_1 = 0.092$, $d_2 = 0.0203$, $T = 0.0822$:

$$\begin{aligned}1 + d_1 d_2 &= 1 + 0.092 \times 0.0203 \approx 1.0019, \\ \frac{1 + d_1 d_2}{2T} &= \frac{1.0019}{2 \times 0.0822} \approx 6.093, \\ b - \frac{1 + d_1 d_2}{2T} &= 0.05 - 6.093 \approx -6.043, \\ \text{Color} &= 0.0310 \times (-6.043) \approx -0.1873.\end{aligned}$$

For a 1-day (0.00274 years) decay, Gamma decreases by $0.1873 \times 0.00274 \approx 0.00051$.

8.3 Use Cases

- **Hedging Horizon:** When Color is large, Gamma erosion is rapid; hedge more often near expiry.
- **Margin Forecasting:** Gamma affects margin; use Color to project margin requirements.

9 Implementing in a Risk System

9.1 Workflow

1. Market data feed: Ingest live S , implied vols, rates, dividends.
2. Batch Greek computation: Compute Γ , GammaP, VommaGamma, Speed, Color daily or on demand.
3. Risk dashboard: Show aggregated exposures: total GammaP by underlying, VommaGamma by volatility bucket, largest Speed values.
4. Alerts: Notify when Gamma or VommaGamma exceed thresholds or when Color signals rapid Gamma decay.
5. Hedge execution: Use Speed and Color to guide size and timing of Delta hedges.

9.2 Sample Python Pseudocode

```
# Given S, K, T, r, q, sigma
import math
```

```
def compute_greeks(S, K, T, r, q, sigma):
    b = r - q
    d1 = (math.log(S/K) + (b + 0.5*sigma**2)*T) / (sigma*math.sqrt(T))
    d2 = d1 - sigma*math.sqrt(T)
    gamma = math.exp(-b*T)/(S*sigma*math.sqrt(2*math.pi*T)) * math.exp(-0.5*d1**2)
    vomma_gamma = gamma * (d1*d2 - 1)/sigma
    speed = -math.exp(-b*T)/(S*S*sigma*math.sqrt(2*math.pi*T)) * math.exp(-0.5*d1**2) * (2 - d1**2)
    color = gamma*(b - (1 + d1*d2)/(2*T))
    gamma_p = 100 * gamma/S
    return {"Gamma": gamma, "GammaP": gamma_p,
            "VommaGamma": vomma_gamma, "Speed": speed, "Color": color}
```

10 Summary and Best Practices

- Compute and monitor Gamma and GammaP daily for all liquid strikes; aggregate by maturity buckets.
- Use VommaGamma to understand how Gamma profiles shift with volatility moves; hedge Vega accordingly.
- Use Speed and Color to forecast hedging needs when spot moves or time passes.
- Incorporate saddlepoint approximations in large-portfolio contexts to save CPU time.
- Validate Greeks by backtesting small price moves to ensure model accuracy in production.